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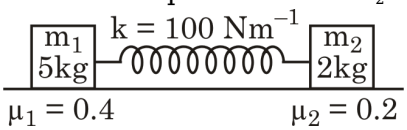
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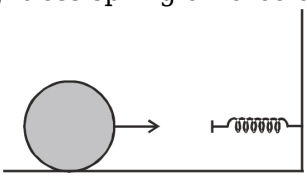
PHYSICS

1) Two blocks m_1 = 5kg and m_2 = 2kg are connected at the two ends of a spring of force constant k = 100 Nm^{-1}. Friction coefficient between m_1 and ground is 0.4, and between m_2 and ground is 0.2. The minimum horizontal velocity v that must be imparted to the m_2 towards right in order to just move m_1 over surface is : (g = 10m/s^2)



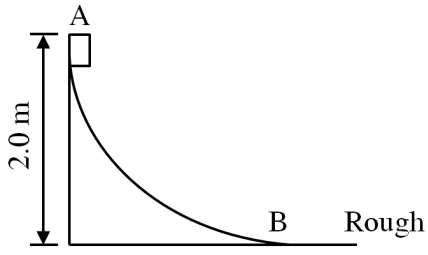
(1) $\sqrt{1.4}$ m/s
(2) $\sqrt{1.8}$ m/s
(3) $\sqrt{2.2}$ m/s
(4) $\sqrt{2.8}$ m/s

2) A mass of 0.5 kg moving with a speed of 1.5 m/s on a horizontal smooth surface, collides with a nearly weightless spring of force constant K = 50 N/m. The maximum compression of the spring would be :-



(1) 0.15 m
(2) 0.12 m
(3) 1.5 m
(4) 0.5 m

3) A block weighing 40 N travels down a smooth fixed curved track AB, joined to a rough horizontal surface (figure). The rough surface has a friction coefficient of 0.10 with the block. If the block starts slipping on the track from a point 2.0 m above the horizontal surface, the horizontal distance it will move on the rough surface is :-



(1) 40 m
(2) 20 m
(3) 15 m
(4) 10 m

4) The rate of doing work by force acting on a particle moving along x-axis depends on position x of

particle and is equal to 2x. The velocity of particle is given by expression :-

(1) $\left[\frac{3x^2}{m}\right]^{1/3}$

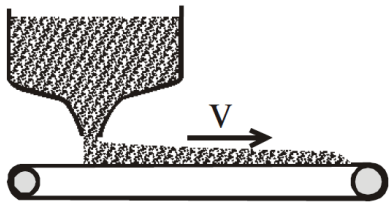
(2) $\left[\frac{3x^2}{2m}\right]^{1/3}$

(3) $\left(\frac{2mx}{9}\right)^{1/2}$

(4) $\left[\frac{mx^2}{3}\right]^{1/2}$

5)

A hopper drops gravel at a rate of 75 kg/sec onto a conveyor belt moving at a constant speed v = 2.2 m/sec. Find power output must the motor have that drives the conveyor belt ?



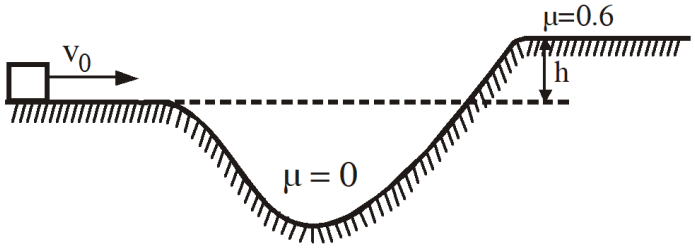
(1) 300 W

(2) 400 W

(3) 263 W

(4) 363 W

6) In the figure, a block slides along a track from one level to a higher level, by moving through an intermediate valley. The track is frictionless untill the block reaches the higher level. There a frictional force stops the block in a distance d. The block's initial speed v_0 is 6 m/s, the height difference h is 1.1m and the coefficient of kinetic friction μ is 0.6. The value of d is



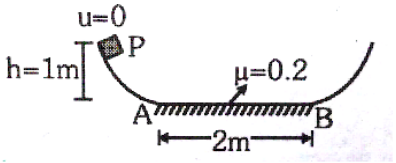
(1) 1.17 m

(2) 1.71 m

(3) 7.11 m

(4) 11.7 m

7) 'The curved portions are smooth and horizontal surface is rough. The block is released from P. At



what distance from A it will stop ?

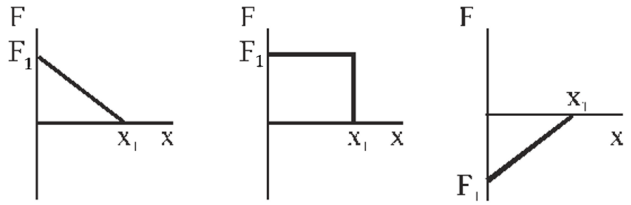
(1) 1 m

(2) 2 m

(3) 3 m

(4) 5 m

8) The graphs below shows the magnitude of the force on a particle as the particle moves along the positive x axis from the origin to $x = x_1$. The force is parallel to the x axis and is conservative. The maximum magnitude F_1 has the same value for all graphs. Rank the situations according to the change in the potential energy associated with the force, least (or most negative) to greatest (or most positive).



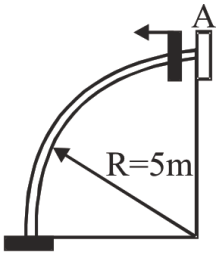
(1) 1,2,3

(2) 1,3,2

(3) 3,2,1

(4) 2,1,3

9) A bead of mass $\frac{1}{2}$ kg starts from rest from A to move in a vertical plane along a smooth fixed quarter ring of radius 5 m, under the action of a constant horizontal force F = 5 N as shown. The speed of bead as it reaches the point B is [Take g = 10 m/s^2] :-



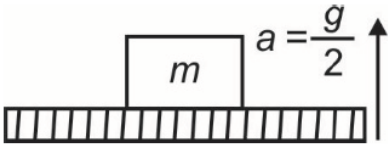
(1) 14.14 m/s

(2) 7.07 m/s

(3) 5 m/s

(4) 25 m/s

10) A block of mass m is kept on a platform which starts from rest with constant acceleration $\frac{g}{2}$ upward, as shown in figure. Work done by normal reaction on block in time t is



(1) $\frac{3mg^2t^2}{8}$

(2) $-\frac{mg^2t^2}{8}$

(3) 0

(4) $\frac{mg^2t^2}{8}$

11) **Statement–I :** A block of mass m starts moving on a rough horizontal surface with a velocity v. It stops due to friction between the block and the surface after moving through a certain distance. The surface is now tilted to an angle of 30° with the horizontal and the same block is made to go up on the surface with the same initial velocity v. The decrease in the mechanical energy in the second situation is smaller than that in the first situation. and
Statement–II : The coefficient of friction between the block and the surface decreases with the increase in the angle of inclination.

(1) statement–I is true, statement–II is true; statement–II is a correct explanation for statement–I

(2) statement–I is true, statement–II is true, statement–II is NOT a correct explanation for statement–I

(3) statement–I is true, statement–II is false

(4) statement–I is false, statement–II is true

12) Power dissipated in an LCR series circuit connected to an a.c. source of emf E is

(1) $\frac{E^2R}{\left[R^2+\left(L\omega-\frac{1}{C\omega}\right)^2\right]}$

(2) $\frac{E^2\sqrt{R^2+\left(L\omega-\frac{1}{C\omega}\right)^2}}{R}$

(3) $\frac{E^2\left[R^2+\left(L\omega-\frac{1}{C\omega}\right)^2\right]}{R}$

(4) $\frac{E^2R}{\sqrt{R^2+\left(L\omega-\frac{1}{C\omega}\right)^2}}$

13) If a current I given by $I_0\sin\left(\omega t-\frac{\pi}{2}\right)$ flows in an ac circuit across which an ac potential of E = E_0 sinωt has been applied, then the power consumption P in the circuit will be

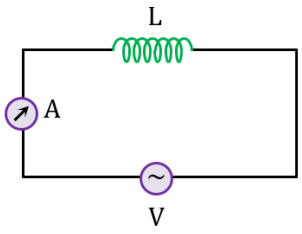
(1) $P=\frac{E_0I_0}{\sqrt{2}}$

(2) $P=\sqrt{2}E_0I_0$

(3) $P=\frac{E_0I_0}{2}$

(4) P = 0

14) In given circuit applied voltage V = $50\sqrt{2}$ sin (100πt) volt and ammeter reading is 2A then



calculate value of L?

(1) $\frac{1}{4\pi}$H

(2) $\frac{1}{2\pi}$H

(3) $\frac{1}{\pi}$H

(4) 2πH

15) In an ac circuit, R = 4Ω, Z = 5Ω, ε_{rms} = 200 V and I_{rms} = 1.5A. Calculate the average power over a full cycle.

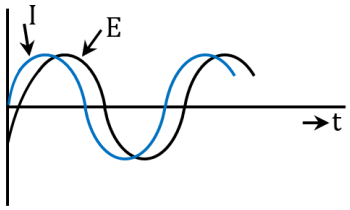
(1) 240 W

(2) 50 W

(3) 40 W

(4) 100 W

16) When an AC source of e.m.f. E = E_0 sin(100t) is connected across as circuit, the phase difference between the e.m.f. E and the current I is observed to be $\frac{\pi}{4}$, as shown in the figure. If the circuit consists possible only of R-C or R-L in series, which of the following combinations is possible



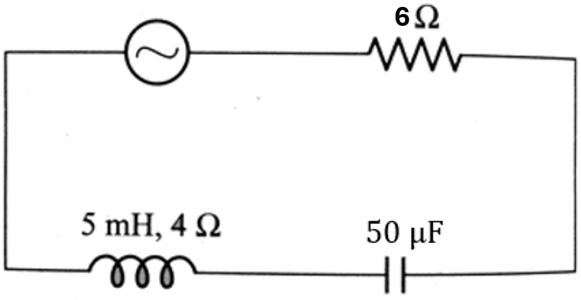
(1) R = 1kΩ, C = 10μF

(2) R = 1kΩ, C = 1μF

(3) R = 1kΩ, L = 10H

(4) R = 1kΩ, L = 1H

17) In the circuit shown below, the ac source has voltage V = 20 cos(ωt) volts with ω = 2000 rad/sec. The amplitude of the current will be to



(1) 2A

(2) 3.3A

(3) $2\sqrt{5}A$

(4) $\sqrt{5}A$

18) **Assertion:** If the frequency of alternating current in an ac circuit consisting of an inductance coil is increased then current gets decreased.
Reason: The current is inversely proportional to frequency of alternating current.

(1) If both assertion and reason are true and the reason is the correct explanation of the assertion.

(2) If both assertion and reason are true but reason is not the correct explanation of the assertion.

(3) If assertion is true but reason is false.

(4) If assertion is false but reason is true.

19) An inductor 20 mH. a capacitor 100 μF and a resistor 50Ω are connected in series across a source of emf, V = 10 sin 314 t. The power loss in the circuit is

(1) 0.79 W

(2) 0.43 W

(3) 2.74 W

(4) 1.13 W

20) In a pure capacitive circuit if a frequency of current is ω then frequency of instantaneous power will be

(1) ω

(2) $\frac{\omega}{2}$

(3) 2ω

(4) $\sqrt{2}\omega$

21) $V = 200\sin\left[\omega t + \frac{\pi}{6}\right]$ and $I = 10\sin\left[\omega t + \frac{\pi}{3}\right]$ calculate power

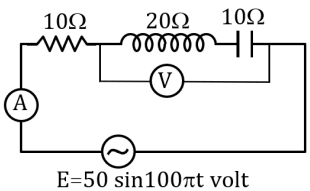
(1) $500\sqrt{2}W$

(2) $500\sqrt{3}W$

(3) $200\sqrt{3}W$

(4) $300\sqrt{3}W$

22) In given circuit find the reading of ammeter and voltmeter.



(1) 26 V

(2) 25 V

(3) 2.5 V

(4) 2.3 V

23) The power factor of an ac circuit is 0.5. What will be the phase difference between the voltage and current in this circuit?

(1) 30°

(2) 45°

(3) 37°

(4) 60°

CHEMISTRY

1) What is the correct IUPAC name of the compound?



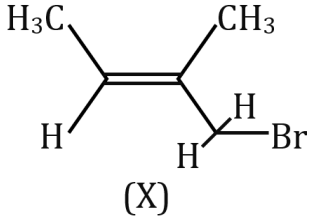
(1) 1-Ethylcyclopent-2-en-3-ol

(2) 4-Ethylcyclopent-2-en-1-ol

(3) 1-Ethyl-4-hydroxycyclopent-2-ene

(4) 4-Ethyl-1-hydroxycyclopent-2-ene

2) Which of the following is the correct IUPAC name of given organic compound (X)?



(X)

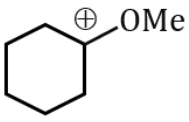
(1) 4-Bromo-3-methylbut-2-ene

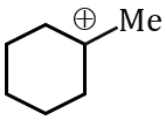
(2) 1-Bromo-2-methylbut-2-ene

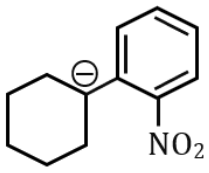
(3) 3-Bromo-3-methylprop-2-ene

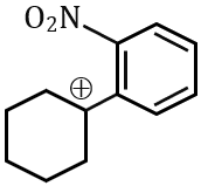
(4) 2-Bromo-2-methylbut-2-ene

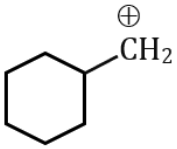
3) In which pairs, the first ion is more stable than the second?

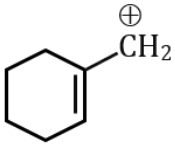
(A) 

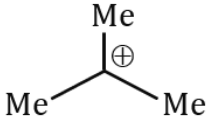
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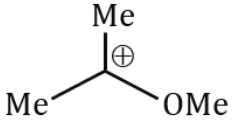
(B) 

&


(C) 

&


(D) 

&


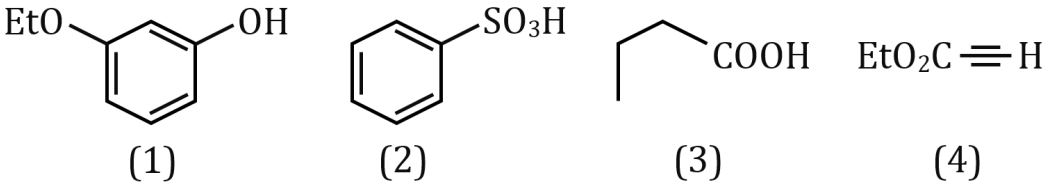
(1) (A) and (B) only

(2) (B) and (D) only

(3) (A) and (C) only

(4) (B) and (C) only

4) The least acidic compound, among the following is:



(1) 1

(2) 2

(3) 3

(4) 4

5) In 3,3-dimethylhex-1-en-4-yne, there are _____sp^3,_____ sp^2 and_____ sp hybridised carbon atoms respectively.

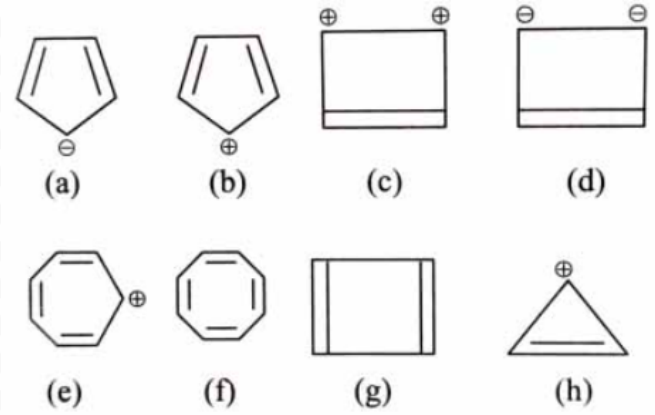
(1) 4, 2, 2

(2) 2, 4, 2

(3) 3, 3, 2

(4) 2, 2, 4

6) Designate whether each of the following compounds is aromatic or not aromatic.



(1) a, c, d, e, h aromatic and b, f, g not aromatic

(2) b, e, f, g aromatic and a, c, d, h not aromatic

(3) a, b, c, d aromatic and e, f, g, h not aromatic

(4) e, g aromatic and a, b, c, d, f, h not aromatic

7) Given below are two statements:

Statement I: Ozonolysis followed by treatment with Zn, H_2O of cis-2-butene gives ethanal.

Statement II: The product obtained by ozonolysis followed by treatment with Zn, H_2O of 3, 6-dimethyloct-4-ene has two carbonyl groups.

In the light of the above statements, choose the correct answer from the options given below

(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are true

8) The compound that will react most readily with gaseous bromine and CCl_4 has the formula

(1) C_4H_{10}

(2) C_4H_8 (Cyclic)

(3) C_3H_6

(4) C_2H_6

9) Identify product [A], [B] and [C] in the following reaction sequence.

$CH_3–C\equiv CH \xrightarrow[H_2]{Pd/C}$ [A] $\xrightarrow[(ii)\,Zn,H_2O]{(i)\,O_3}$ [B] + [C]

(1) [A]: CH_3-CH=CH_2, [B]: CH_3CHO, [C]: CH_3CH_2OH

(2) [A]: CH_3-CH=CH_2, [B]: CH_3CHO, [C]: HCHO

(3) [A]: $CH_3CH_2CH_3$, [B]: CH_3CHO, [C]: HCHO

(4) $[A] : CH_2 = CH_2, [B] : H_3C - \overset{O}{\overset{\|}{C}} - CH_3\ [C] : HCHO$

10) $C_2H_5Br \xrightarrow{alc.\,KOH}$ (A) $\xrightarrow[CCl_4]{Br_2}$ (B)

IUPAC name of (B) formed in the above reaction is :

(1) ethylene bromide

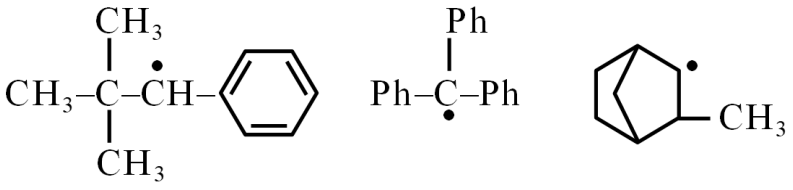
(2) 1,2-dibromo ethane

(3) 1,2-dibromo ethene

(4) None of these

11)

Consider the following compounds



(I)

(II)

(III)

Hyperconjugation occurs in :-

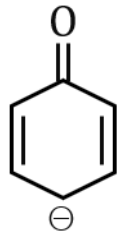
(1) II only

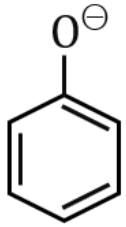
(2) III only

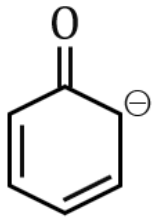
(3) I and III

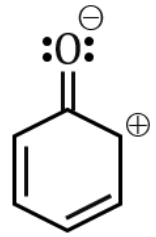
(4) I only

12) Which of the following is not a valid resonance structures of phenoxide ion ?

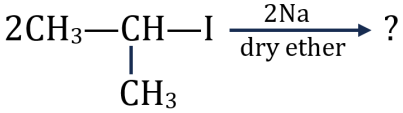
(1) 

(2) 

(3) 

(4) 

13) Identify the product of the following reaction:-



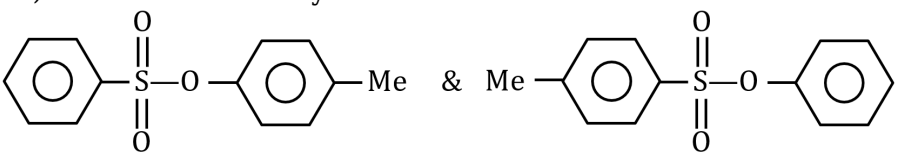
(1) Propane

(2) 2, 3–dimethyl butane

(3) 1, 2–dimethyl butane

(4) 2–Butene

14) Isomerism shown by



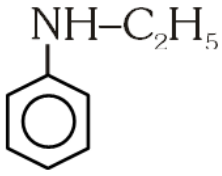
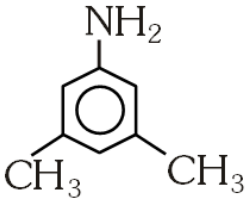
(1) Chain isomerism

(2) Positional isomerism

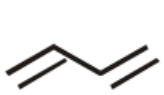
(3) Metamerism

(4) Functional group isomerism

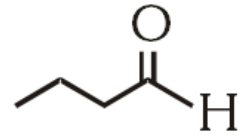
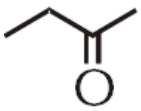
15) Identify relation that is incorrect

(1)  and 

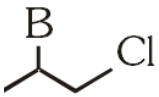
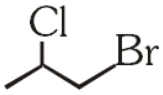
functional isomers

(2)  and 

Ring chain isomers

(3)  and 

Metamers

(4)  and 

Position isomers

16) Most suitable method of separation of 1 : 1 mixture of ortho and para nitrophenol is :-

(1) Chromatography

(2) Crystallisation

(3) Sublimation

(4) Steam distillation.

17) Statement- I : Angle required to get maximum stable conformation from minimum stable in n Butane is 180°.
Statement- II : Maximum stable conformation of n Butane is called as Gauche form.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

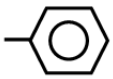
(3) Statement I is true but Statement II is false.

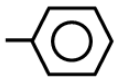
(4) Statement I is false but Statement II is true.

18) Correct order of –I effect ?

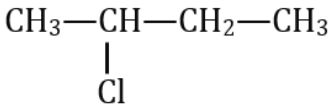
(1) $-CN > -NO_2 > -X$

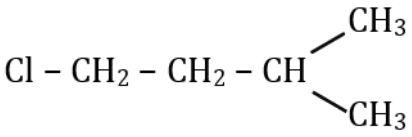
(2) $-NO_2 > -OCH_3 > -C\equiv N$

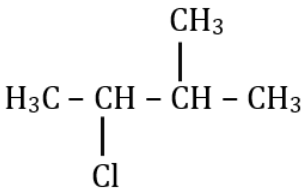
(3) $-NO_2 > -\overset{\oplus}{N}H_3 >$ 

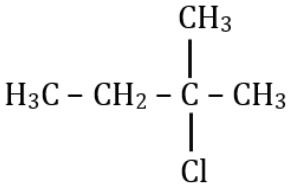
(4) $-\overset{\oplus}{N}H_3 > -C\equiv N >$ 

19) An alkene A on reaction with O_3 & Zn – H_2O gives propanone & ethanal in equimolar ratio addition of HCl to alkene A gives B as a minor product. The structure of product B.

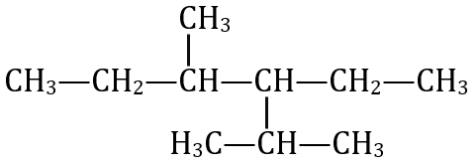
(1) 

(2) 

(3) 

(4) 

20) The IUPAC name for the given structure is :-



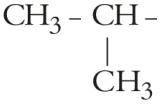
(1) 3–Isopropyl–4–methylhexane

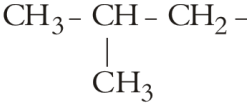
(2) 4–Isopropyl–3–methylhexane

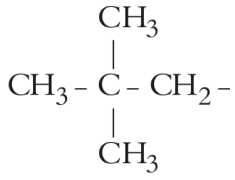
(3) 3–Ethyl–2,4–dimethylhexane

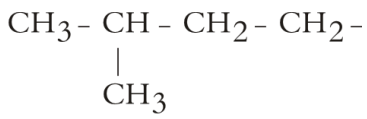
(4) 2–Ethyl–3–isopropylpentane

21) Which name is incorrect among the following alkyl groups?

(1)  iso propyl

(2)  iso butyl

(3)  t-butyl

(4)  iso pentyl

22) **Assertion (A) :** acetylene can react with $NaNH_2$ but does not react with NaOH
Reason (R) : Acetylene is more acidic than alcohol

(1) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A).**

(2) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A).**

(3) **(A)** is correct but **(R)** is not correct.

(4) **(A)** is not correct but **(R)** is correct.

BIOLOGY

1) Presence of XX or XY chromosomes in zygote depends on -

(1) The sperm carrying X chromosome fertilized the ovum.

(2) The sperm carrying Y chromosome fertilized the ovum.

(3) The sperm without any chromosome fertilized the ovum.

(4) The sperm carrying X or Y chromosomes fertilized the ovum.

2) Second meiotic division in secondary oocyte results in the formation of -

(1) First polar body and a diploid ovum.

(2) First polar body and a haploid ovum.

(3) Second polar body and a diploid ovum.

(4) Second polar body and a haploid ovum.

3) The sperm and the egg make different contributions to zygote.
Which of the following statements about their contributions are true?
(i) Sperm contributes most of the mitochondria.
(ii) Egg contributes most of the cytoplasm.
(iii) Both sperm and egg contribute haploid nucleus.
(iv) Both sperm and egg contribute centrioles.

(1) (i) and (ii)

(2) (ii) and (iii)

(3) (iii) and (iv)

(4) All of these

4)

Read the following statements (i to iv) and answer the following question.
(i) Each testes has highly coiled 250 compartments called seminiferous tubules.
(ii) Erection of the penis due to presence of special tissues facilitates insemination.
(iii) Immunologically competent cells are also present in the interstitial spaces of seminiferous tubules.
(iv) Testes lie outside the abdominal cavity in a thin pouch like skin called scrotum.
(v) Bulbourethral gland is a single accessory gland.
How many of the above statements are incorrect?

(1) (i), (ii) and (iii)

(2) (iii) and (v)

(3) (i) and (v)

(4) (ii), (iv) and (v)

5)

Read the following statements (i to v) and answer the following question.
(i) This structure is also called womb.
(ii) Its shape is like an inverted pear.
(iii) The process of fertilization takes place in this structure.
(iv) The wall of this structure has three layers of tissue.
(v) It secretes several steroid hormones.
Identify the correct characteristics feature regarding uterus from the above statements.

(1) (i) and (iv)

(2) (iii) and (v)

(3) (i), (ii) and (iv)

(4) All the five statements

6) Which of the following statements regarding mammary gland is incorrect?

(1) They are paired glandular structure that lies over the pectoral muscles.

(2) Each gland has 100 – 500 lobulated milk glands each having a number of lobules containing number of alveoli.

(3) The cells of alveoli secrete milk which is stored in the cavity of the alveoli.

(4) Each milk gland or lobules has lactiferous ducts that drain into openings in the nipple.

7)

Select the correct statements regarding oogenesis.
(i) It is initiated during the embryonic development stage when millions of oogonia are formed within each ovary.
(ii) Graafian follicle releases primary oocyte from the ovary by ovulation.
(iii) At puberty only 60,000 – 80,000 primary follicles are left in each ovary.
(iv) Secondary oocyte within tertiary follicles grows in size and completes its second meiotic division.

(1) (i), (ii) and (iii)

(2) (i) and (iii)

(3) (ii) and (iv)

(4) All the four statements

8)

Which of the following is required for the increased production of estrogen, progestogens, cortisol, prolactin and thyroxine etc. in the maternal blood?
(i) Metabolic changes in the mother.
(ii) Maintenance of pregnancy.
(iii) Supporting the foetal growth.
(iv) Destruction of Graafian follicle.

(1) (iii) and (iv)

(2) (i), (iv) and (v)

(3) (i), (ii) and (iii)

(4) All the four statements

9)

Which of the following statement regarding female reproductive system is (are) correct?
(i) Myometrium undergoes strong contraction at the time of delivery of baby.
(ii) Ovary is secondary female sex organ which produces female gamete and steroid hormones.
(iii) Ovarian stroma is divided into two zones: inner cortex and outer medulla.
(iv) Infundibulum possess finger like projections which help in collection of ovum after the release of secondary oocyte.
(v) A functional mammary gland is the characteristics of all the mammals (including male and female).

(1) (i) and (iv)

(2) (i), (ii), (iii) and (v)

(3) (iii), (iv) and (v)

(4) All the five statements

10)

Read the following statements (i to v) and answer the question.
(i) It produces several hormones like hCG, hPL, estrogens, progestogens etc.
(ii) It differentiates into three embryonic membranes – ectoderm, endoderm and mesoderm.
(iii)It undergoes mitotic division.
(iv)It is the organ, formed in the lining of the uterus by the union of the uterine mucous membrane with the membranes of the foetus.
(v) It develops at a point of implantation and providing oxygen and nutrients for the foetus and transfer of waste products from the foetal to the maternal blood circulation.
Identify the correct characteristics feature regarding placenta from the above statements.

(1) (iii) and (v)

(2) (i), (iv) and (v)

(3) (i), (ii) and (iv)

(4) All the four statements

11) Which of the following statements regarding parturition is incorrect?

(1) Prolactin induces uterine contraction.
(2) It is induced by neuroendocrine mechanism.
(3) Uterine contraction leads to expulsion of baby through the birth canal.
(4) Oxytocin plays an important role in the contraction of uterus

12) **Assertion :** During fertilization only head of spermatozoa enters egg.
Reason : If several spermatozoa hit the egg at same time, all can enter the egg.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

13) **Assertion :** Corpus luteum degenerates in the absence of fertilization.
Reason : Progesterone level decreases.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

14) **Assertion :** Clitoris is not remnant of penis in females.
Reason : It also have blood supply and erectile tissue.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

15) **Assertion :** Mammalian ova have plenty amount of yolk.
Reason : The eggs of mammal covered with shell.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.
(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.
(3) Assertion is true but Reason is false.
(4) Both Assertion and Reason are false.

16) **Assertion :** Interstitial cell is present in the region outside the seminiferous tubule called interstitial spaces.
Reason : Interstitial cells provide nutrition to the Sertoli cells.

(1) Both Assertion and Reason are true and the Reason is the correct explanation of the Assertion.

(2) Both Assertion and Reason are true but the Reason is not the correct explanation of the Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

17) In the given columns, column I contain structures of male reproductive system and column II contains its feature. Select the correct match from the options given below.

Column I		Column II	
A.	Seminiferous tubule	I.	Network of seminiferous tubule
B.	Rete testis	II.	Developed Secondary sexual characters
C.	Leydig cells	III.	Meiosis and sperm formation occurs
D.	Prepuce / foreskin	IV.	Place of implantation
		V.	Terminal skin of penis

(1) A – I; B – II; C – III; D – V

(2) A – III; B – I; C – II; D – V

(3) A – III; B – I; C – IV; D – II

(4) A – II; B – IV; C – III; D – V

18) Which of the following pair is incorrectly matched?

(1) Leydig cells – Testosterone

(2) Spermatogenesis – Seminiferous tubules

(3) Male reproductive system – Pelvis region

(4) Spermatogonia – Mitotic division

19) In the given columns, column-I contain structures of female reproductive system and column-II contain its feature.
Select the correct match from the option given below.

Column I		Column II	
A.	Ampulla	I.	It undergoes cyclical changes during menstrual cycle.
B.	Labia majora	II.	It helps in collection of ovum after ovulation.
C.	Oviduct	III.	Wider part of fallopian tube where fusion of male and female gametes takes place.
D.	Fimbriae	IV.	Larger hairy folds which extend down from the mons pubis and surrounds the vaginal opening.
E.	Endometrium	V.	Also called fallopian tubes, which extend from the periphery of each ovary to the womb.

(1) A – I; B – II; C – III; D – V; E – IV

(2) A – III; B – I; C – II; D – V; E – IV

(3) A – III; B – IV; C – V; D – II; E – I

(4) A – II; B – IV; C – III; D – V; E – I

20) Match the column-I with column-II and select the correct option.

Match from the options given below.

Column I		Column II	
A.	Primary oocyte	I.	It is formed when oogonia starts division and temporarily arrested at prophase of meiosis I.
B.	Secondary oocyte	II.	A large haploid cell which retains bulk of nutrient rich cytoplasm of the primary oocyte.
C.	Primary follicle	III.	A large number of these degenerate during the phase from puberty to birth.
D.	Oogonia	IV.	Gamete mother cell.
E.	Secondary follicle	V.	Surrounded by more layers of granulosa cells and a new theca.
F.	Graafian follicle	VI.	Rupture to release ovum from the ovary.

(1) A – I; B – II; C – III; D – IV; E – V; F – VI

(2) A – III; B – I; C – IV; D – II; E – V; F – VI

(3) A – VI; B – IV; C – V; D – II; E – I; F – III

(4) A – II; B – IV; C – III; D – V; E – I; F – VI

21) In the given columns, column-I contain various phases of menstrual cycle and column -II contain its features.
Select the correct match from the options given below.

Column I		Column II	
A.	Menstrual phase	I.	Breakdown of endometrial lining of uterus along with its blood vessels which form liquid that comes out of vagina
B.	Luteal phase	II.	A temporary endocrine is formed and secretes a hormone which maintains endometrium and implantation of fertilized ovum events of pregnancy.
C.	Follicular phase	III.	Secretion of luteinizing hormone at its maximum level and induces breakdown of mature follicle to release the female gamete.
D.	Ovulatory phase	IV.	Formation of mature Graafian follicle and regeneration of endometrium of uterus.

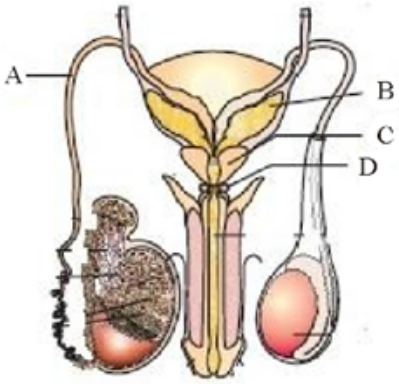
(1) A – IV; B – II; C – III; D – I

(2) A – III; B – I; C – II; D – IV

(3) A – III; B – I; C – IV; D – II

(4) A – I; B – II; C – IV; D – III

22) Given below is a diagrammatic sketch of a portion of human male reproductive system. Select the correct set of the names of the parts marked as A, B, C, and D respectively.



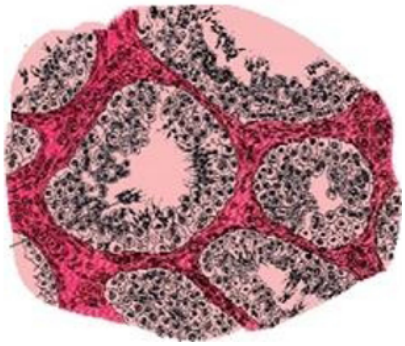
(1) A–Ureter, B–Seminal vesicle, C–Prostate, D–Bulbourethral gland

(2) A–Ureter, B–Prostate, C–Seminal vesicle, D–Bulbourethral gland

(3) A–Vas deferens, B–Seminal vesicle, C–Prostate, D–Bulbourethral gland

(4) A–Vas deferens, B–Seminal vesicle, C–Bulbourethral gland, D–Prostate deferens

23) Identify the figure, (A) whose sectional view is given below and match with its characteristics (B)



and its location (C).

	A	B	C
(1)	Graafian follicle	Involved in the formation of ovum	Ovary
(2)	Seminiferous tubule	Involved in the formation of sperm	Testis
(3)	Ovum surrounded by sperm	Process of fertilization	Graafian follicle
(4)	Mammary gland	Involved in milk secretion	Female reproductive system

(1) 1

(2) 2

(3) 3

(4) 4

24) **Assertion:** In most angiosperms, pollen grains are shed at 3-celled stage.
Reason: Generative cell divides before pollen release.

(1) Both A and R are true and R is the correct explanation

(2) Both A and R are true but R is not the correct explanation

(3) A is true but R is false

(4) Both A and R are false

25) **Assertion:** The embryo sac in angiosperms is 7-nucleate but 7-celled.

Reason: The two polar nuclei remain free in the central cell.

(1) Both A and R are true and R is the correct explanation
(2) Both A and R are true but R is not the correct explanation
(3) A is true but R is false
(4) A is false but R is true

26) Match the flower parts with their description:

	Column A		**Column B**
A.	Hilum	P.	Junction between ovule and funicle
B.	Funicle	Q.	Stalk of ovule
C.	Micropyle	R.	Opening in integuments
D.	Chalaza	S.	Basal part of ovule

(1) A-P, B-Q, C-R, D-S
(2) A-Q, B-P, C-S, D-R
(3) A-S, B-R, C-P, D-Q
(4) A-P, B-R, C-Q, D-S

27) Match the following components with their functions:

	Column A		**Column B**
A.	Tapetum	P.	Nutrition of microspores
B.	Exine	Q.	Protection and design
C.	Filiform apparatus	R.	Pollen tube guidance
D.	Germ pore	S.	Site of pollen germination

(1) A-P, B-Q, C-R, D-S
(2) A-R, B-S, C-Q, D-P
(3) A-Q, B-R, C-P, D-S
(4) A-P, B-R, C-Q, D-S

28) Match the developmental stages with structures:

	Column A		**Column B**
A.	MMC	P.	Undergoes meiosis
B.	Dyad	Q.	Two megaspore cells
C.	Tetrad	R.	Four haploid megaspores
D.	Functional megaspore	S.	Develops embryo sac

(1) A-P, B-Q, C-R, D-S
(2) A-R, B-S, C-Q, D-P

(3) A-Q, B-R, C-P, D-S

(4) A-P, B-R, C-Q, D-S

29) Arrange the stages of microsporogenesis in correct order:
A. Pollen mother cell
B. Meiosis
C. Microspore tetrad
D. Pollen grain

(1) A, B, C, D

(2) B, A, C, D

(3) A, C, B, D

(4) C, B, A, D

30) Arrange the stages in development of embryo sac from functional megaspore:
A. Functional megaspore
B. 2-nucleate stage
C. 4-nucleate stage
D. 8-nucleate stage

(1) A, B, C, D

(2) B, C, A, D

(3) D, C, B, A

(4) A, C, B, D

31) Identify the incorrect statement:

(1) Tapetum is the outermost wall layer in anther

(2) Exine is made of sporopollenin

(3) Germ pores lack sporopollenin

(4) Intine is made of cellulose and pectin

32) Which of the following is a **correct** statement?

(1) Nucellus lies outside the integuments

(2) Each ovule contains many embryo sacs

(3) Hilum connects ovule and placenta

(4) Megaspore mother cell is diploid

33) Which is **incorrect** about pollen viability?

(1) Pollen of wheat is viable for 30 minutes

(2) Viability depends on temperature and humidity

(3) Pollen can be stored in liquid nitrogen

(4) Rosaceae pollen loses viability quickly

34) Which of the following is **correct**?

(1) Ovary serves as landing platform for pollen

(2) Style is site of fertilization

(3) Micropyle allows pollen tube entry

(4) Funicle is basal part of embryo sac

35) Identify the **incorrect** statement regarding the embryo sac:

(1) Antipodals are at chalazal end

(2) Synergids have filiform apparatus

(3) Central cell contains one polar nucleus

(4) It is 8-nucleate but 7-celled

36) Arrange the following steps in the correct chronological order during the process of fertilisation in angiosperms:
A. Pollen tube enters the synergid
B. Generative cell divides into two male gametes
C. Entry of pollen on stigma
D. Discharge of male gametes into the embryo sac

(1) C, A, B, D

(2) C, B, A, D

(3) C, A, D, B

(4) C, B, D, A

37) Which of the following statements are not true for cross-pollination?

(1) Unisexual flowers promotes cross pollination

(2) Can be facilitated by agents such as wind and insects

(3) Promotes homozygosity in plant populations

(4) Enhances genetic variability in offspring

38) Match the pollination types with their correct examples:

	Column A (Pollination type)		**Column B (Example)**
A.	Entomophily	P.	*Salvia*
B.	Anemophily	Q.	*Maize*
C.	Hydrophily (epihydrophily)	R.	*Vallisneria*
D.	Ornithophily	S.	*Bignonia* (bird-pollinated flower)

(1) A-P, B-Q, C-R, D-S

(2) A-R, B-P, C-S, D-Q

(3) A-S, B-Q, C-P, D-R

(4) A-Q, B-R, C-S, D-P

39) A researcher finds that flowers of an unknown plant species are small, colourless, lack nectar,

and produce large quantities of dry pollen. Which conclusion is most scientifically justified?

(1) The plant is likely pollinated by insects
(2) The plant is likely pollinated by birds
(3) The plant is likely pollinated by wind
(4) The plant is likely self-pollinating only

40) Match the floral adaptation (Column A) with its corresponding pollination mechanism (Column B):

	Column A		Column B
A.	Sticky pollen grains	P.	Entomophily
B.	Light pollen	Q.	Anemophily
C.	Coloured petals	R.	Insect pollination
D.	Cleistogamy	S.	Autogamy

(1) A-P, B-Q, C-R, D-S
(2) A-Q, B-P, C-R, D-S
(3) A-R, B-P, C-Q, D-S
(4) A-R, B-Q, C-P, D-Q

41) Match the following flower types with the correct description:

	Column A		Column B
A.	Chasmogamous	P.	Open flowers, allow both types
B.	Cleistogamous	Q.	Closed flowers, obligate autogamy
C.	Geitonogamous	R.	Self-pollination between flowers of same plant
D.	Xenogamous	S.	Pollination between flowers of different plants

(1) A-P, B-Q, C-R, D-S
(2) A-Q, B-P, C-S, D-R
(3) A-P, B-R, C-Q, D-S
(4) A-R, B-Q, C-P, D-S

42) In angiosperms, if triple fusion does not occur, which of the following structures will not form?

(1) Embryo
(2) Endosperm
(3) Seed coat
(4) Ovule

43) Arrange the following post-fertilisation events in correct sequence:
A. Development of endosperm
B. Formation of zygote
C. Differentiation of embryo

D. Formation of seed coat

(1) B, A, C, D
(2) A, B, C, D
(3) B, C, A, D
(4) D, C, B, A

44) Match the following seed parts with their origin:

	Column A		**Column B**
A.	Endosperm	P.	Primary endosperm nucleus
B.	Embryo	Q.	Zygote
C.	Seed coat	R.	Integuments
D.	Pericarp	S.	Ovary wall

(1) A-P, B-Q, C-R, D-S
(2) A-R, B-P, C-S, D-Q
(3) A-Q, B-P, C-R, D-S
(4) A-P, B-R, C-Q, D-S

45) Which of the following statements is **incorrect**?

(1) Apomixis is a form of asexual reproduction
(2) Seeds formed via apomixis show genetic variability
(3) Apomixis can occur through the development of an embryo from a nucellar cell
(4) Polyembryony may occur due to multiple embryos in a single ovule

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	1	2	1	4	1	1	4	1	1	3	1	4	1	1	1	1	1	1	3
Q.	21	22	23																	
A.	2	2	4																	

CHEMISTRY

Q.	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43
A.	2	2	1	4	1	1	3	3	2	2	2	4	2	3	3	4	3	4	3	3
Q.	44	45																		
A.	3	3																		

BIOLOGY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	4	2	3	3	2	2	3	1	2	1	3	2	2	4	3	2	4	3	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	2	4	4	1	1	1	1	1	1	4	4	3	3	2	3	1	3	1
Q.	86	87	88	89	90															
A.	1	2	1	1	2															

SOLUTIONS

PHYSICS

1) **Explain the Question :** We have to calculate the minimum horizontal velocity v that must be imparted to m_2 towards right to just move m_1.

Concept :

This question is based on work energy theorem.

Solution :

At the moment m_2 stops, extension in the spring must be able to produce enough force to move m_1 or

$kx = \mu_1 m_1 g \Rightarrow x = \frac{.4 \times 5 \times 10}{100} = 20\text{cm}.$

As it is equal to displacement of m_2 also, applying work-energy theorem on m_2

$0 - \frac{1}{2} m_2 v^2 = W_s + W_f = -\frac{1}{2} kx^2 - \mu_2 m_2 g.\ x$

$\Rightarrow \frac{1}{2}(2)\, v^2 = \frac{1}{2} \times 100 \times 0.2 \times 0.2 + 0.2 \times 2 \times 10 \times 0.2 \quad v^2 = 2 + 0.8 \Rightarrow v = \sqrt{2.8}\ \text{m/s}$

Final Answer : (4)

2) Loss of KE = Gain in elastic PE

$$\frac{1}{2}mv^2 = \frac{1}{2}Kx^2$$

or $0.5 \times (1.5)^2 = 50 \times x^2$

$$x^2 = \frac{0.5 \times (1.5)^2}{50} = \frac{(1.5)^2}{100}$$

$$x = \frac{1.5}{10} = 0.15 \text{ m}$$

3) $W_{mg} + W_N + W_{fr} = \frac{1}{2} m \left(v_f^2 - v_i^2\right)$

$mgh + W_{fr} = 0 - 0$

$W_{fr} = -mgh$

$-\mu_K NS = -mgh$

$-\mu_K mgS = -mgh$

$S = \frac{h}{\mu_k} = \frac{2}{10} \Rightarrow \frac{2 \times 100}{10} = 20 \text{ m}$

4) $P = 2x$

$\Rightarrow F.v = 2x \Rightarrow mav = 2x$

$\Rightarrow m.\frac{vdv}{dx}.v = 2x$

$\Rightarrow v^2 dv = \frac{2}{m} x dx$

on integrating both sides

$\Rightarrow \frac{v^3}{3} = \frac{2}{m}\frac{x^2}{2}$

$\Rightarrow v = \left[\frac{3x^2}{m}\right]^{1/3}$

Hence option (1)

5) $F = \frac{dp}{dt} = m\frac{dv}{dt} + v\frac{dm}{dt}$

$f = m(0) + v\frac{dm}{dt}$

= 2.2 (75) = 165 N

P = FV = 165 (2.2) = 363 W

6)

Question Explanation : We need to find the distance d over which friction stops the block on the upper level.

Concept : Use the work Energy Theorem: total work by gravity and friction equals the change in kinetic energy.

Solution:

Initial KE : $\frac{1}{2}mv_0^2$

Work by gravity = – mgh

work by friction = –μmgd

Final KE : 0

Apply work energy :

$-mgh - \mu mgd = 0 - \frac{1}{2}mv_0^2$

$gh + \mu gd = \frac{1}{2}v_0^2$

Calculation

10(1.1) + 0.6(10) d = 18

11 + 6d = 18

6d = 7 ⇒ d = 7/6

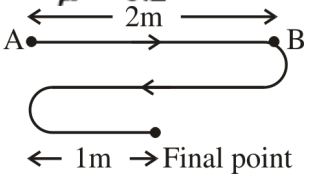
Final Answer : option (1)

7)

By WET

mgh – μmg = 0

$\ell = \frac{h}{\mu} = \frac{1}{0.2} = 5m$



8) $U_B - U_A = \Delta U = -\int_A^B \vec{F}.d\vec{s}$

ΔU = area under the F-x graph

9) From work-energy theorem

$W_F + W_{mg} + W_N = \frac{1}{2} mv^2$

$F \times R + mgR + 0 = \frac{1}{2} mv^2$

$5 \times 5 + \frac{1}{2} \times 10 \times 5 = \frac{1}{2} \times \frac{1}{2} \times v^2$

$\therefore v = \sqrt{200} = 14.14$ m/s

$$N - mg = \frac{mg}{2}$$

$$N = \frac{3mg}{2}$$

$$S = \frac{1}{2}\left(\frac{g}{2}\right)t^2$$

$$W = \frac{3mg}{2} \times \frac{g}{4} t^2$$

10) $$W = \frac{3mg^2}{8} t^2.$$

11) In horizontal plane Kinetic Energy of the block is completely converted into heat due to Friction but in the case of inclined plane some part of this Kinetic Energy is also convert into gravitational Potential Energy. So decrease in the mechanical energy in second situation is smaller than that in the first situation. So statement-1 is correct. Cofficient of Friction does not depends on normal reaction, In II case normal reaction changes with inclination but not cofficient of friction so this statement is wrong.

12) **Explaining :** Power dissipated in an LCR series circuit connected to an a.c. source of emf E is

Concept : This question is based on $P = V_{rms}\, I_{rms} \cos\phi$.

$$P = E_{rms}\ \ i_{rms} \cos\phi = \frac{E^2R}{Z^2} = \frac{E^2R}{\left[R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2\right]}.$$

Solution :

Final Answer : (1)

13) **Explaining :** If a current I given by $I_0 \sin\left(\omega t - \frac{\pi}{2}\right)$ flows in an ac circuit across which an ac potential of $E = E_0 \sin\omega t$ has been applied, then the power consumption P in the circuit will

be
Concept : This question is based on P = VI cosϕ.
Solution : Phase angle ϕ = 90°, so power P = Vi cosϕ = 0
Final Answer : (4)

14) **Explaining :** In given circuit applied voltage V = $50\sqrt{2}$ sin (100πt) volt and ammeter reading is 2A then calculate value of L?
Concept : This question is based on $X_L = L\omega$.
Solution : $V_{rms} = I_{rms} X_L$ ∵ Reading of ammeter = I_{rms}

$$X_L = \frac{V_{rms}}{I_{rms}} = \frac{V_0}{\sqrt{2}\, I_{rms}} = \frac{50\sqrt{2}}{\sqrt{2} \times 2} = 25\ \Omega$$

$$L = \frac{X_L}{\omega} = \frac{25}{100\pi} = \frac{1}{4\pi} H$$

Final Answer : (1)

15) **Explaining :** In an ac circuit, R = 4Ω, Z = 5Ω, ε_{rms} = 200 V and I_{rms} = 1.5A. Calculate the average power over a full cycle.
Concept : This question is based on P = VIcosϕ.
Solution : $P_{av} = \varepsilon_{rms}\ I_{rms} \cos\phi = \varepsilon_{rms}\ I_{rms}\ (R/Z)$
= (200 V)(1.5 A)(4/5) = 240 W
Final Answer : (1)

16) **Explaining :** When an AC source of e.m.f. E = E_0 sin(100t) is connected across as circuit, the phase difference between the e.m.f. E and the current I is observed to be $\frac{\pi}{4}$, as shown in the figure. If the circuit consists possible only of R-C or R-L in series, which of the following combinations is possible
Concept : This question is based on RC circuit.
Solution :

Figure given in the questions shows that current I leads the emf E by a phase angle $\frac{\pi}{4}$.
Therefore, the circuit can be R-C circuit alone.

$$\tan\phi = \frac{X_C}{R} = \frac{1}{\omega CR} \qquad \left[\because X_C = \frac{1}{\omega C}\right]$$

$$\tan\frac{\pi}{4} = \frac{1}{\omega CR}$$

$$1 = \frac{1}{\omega CR} \qquad \text{...(i)}$$

Compare E = E_0 sin(100t) with E = E_0 sin ωt, we get
ω = 100 rad s^{-1}

$$\therefore CR = \frac{1}{\omega} = \frac{1}{100} \qquad \text{[Using (i)]}$$

In all option R = 1 KΩ so put R = 1 KΩ & find C
When R = 1kΩ = 10^3Ω

$$C = \frac{1}{10^5} = 10^{-5}\ F = 10\mu F$$

Final Answer : (1)

17) **Explaining :** In the circuit shown below, the ac source has voltage V = 20 cos(ωt) volts with ω = 2000 rad/sec. The amplitude of the current will be to

Concept : This question is based on $Z = \sqrt{R^2 + (X_L - X_C)^2}$.

Solution : R = 6 + 4 = 10Ω

$X_L = \omega L = 2000 \times 5 \times 10^{-3} = 10\Omega$

$$X_C = \frac{1}{\omega C} = \frac{1}{2000 \times 50 \times 10^{-6}} = 10\Omega$$

$$\therefore Z = \sqrt{R^2 + (X_L - X_C)^2} = 10\Omega$$

Amplitude of current = $i_0 = \frac{V_0}{Z} = \frac{20}{10} = 2A$

Final Answer : (1)

18) **Explaining : Assertion:** If the frequency of alternating current in an ac circuit consisting of an inductance coil is increased then current gets decreased.

Reason: The current is inversely proportional to frequency of alternating current.

Concept : This question is based on $I = \frac{V}{X_L}$.

Solution : When frequency of alternating current is increased, the effective resistance of the inductive coil increases. Current ($X_L = \omega L = 2\pi fL$) in the circuit containing inductor is given by $I = \frac{V}{X_L} = \frac{V}{2\pi fL}$. As inductive resistance of the inductor increases, current in the circuit decreases.

Final Answer : (1)

19) **Explaining :** An inductor 20 mH. a capacitor 100 μF and a resistor 50Ω are connected in series across a source of emf, V = 10 sin 314 t. The power loss in the circuit is

Concept : This question is based on $P = V_{rms}\, I_{rms} \cos\phi$.

Solution : $V_0 = 10V$, ω = 314 rad/s

$P = V_{rms} \quad i_{rms} \cos\phi$

$$= V_{rms}\left(\frac{V_{rms}}{Z}\right)\left(\frac{R}{Z}\right) = \frac{(V_{rms})^2 R}{Z^2}$$

$X_L = \omega L = (314)\,(20 \times 10^{-3}) = 6.280$

$$X_C = \frac{1}{\omega C} = \frac{1}{314 \times 100 \times 10^{-6}} = 31.84\Omega$$

R = 50Ω

$$Z = \sqrt{(X_C - X_L)^2 + R^2}$$

$$= \sqrt{(31.84 - 6.28)^2 + (50)^2} = 56\Omega$$

$$\Rightarrow P = \frac{\left(\frac{10}{\sqrt{2}}\right)^2 \times 50}{(56)^2} = 0.79W$$

Final Answer : (1)

20) **Explaining :** In a pure capacitive circuit if a frequency of current is ω then frequency of instantaneous power will be

Concept : This question is based on P = VI.

Solution : $V = V^0 \sin\omega t$, $I = I^0 \sin(\omega t + \frac{\pi}{2}) = I^0 \cos\omega t$

$p = V \times I$

$p = V_0 I_0 \sin\omega t \cos\omega t$

$p = \frac{V_0 I_0}{2} \sin(2\omega t)$

Final Answer : (3)

21) **Explaining :** $V = 200\sin\left[\omega t + \frac{\pi}{6}\right]$ and $I = 10\sin\left[\omega t + \frac{\pi}{3}\right]$ calculate power

Concept : This question is based on $P = V_{rms}\, I_{rms} \cos\phi$.

Solution : $\phi = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6} = 30^\circ$

$p = V_{rms}\, I_{rms} \cos 30°$

$= \frac{200}{\sqrt{2}} \times \frac{10}{\sqrt{2}} \cos 30^\circ$

$= 500\sqrt{3}W$

Final Answer : (2)

22) **Explaining :** In given circuit find the reading of ammeter and voltmeter.

Concept : This question is based on $V = V_L - V_C$.

Solution : $I = \frac{V}{Z} = \frac{50}{\sqrt{2} \times 10\sqrt{2}} = 2.5A$

$X_L - X_C = 10\Omega$

Potential drop at combination of L & C

$= i(X_L - X_C) = 2.5 \times 10 = 25$ V

Final Answer : (2)

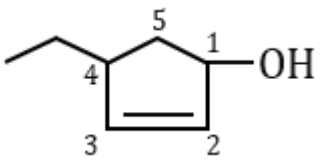
23) **Explaining :** The power factor of an ac circuit is 0.5. What will be the phase difference between the voltage and current in this circuit?

Concept : This question is based on phase difference, $\phi = \cos^{-1}(0.5)$.

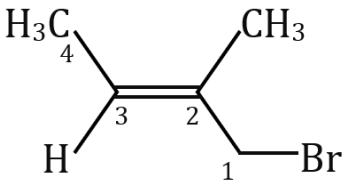
Solution : As power factor, $\cos\phi = 0.5$, $\phi = 60°$.

Final Answer : (4)

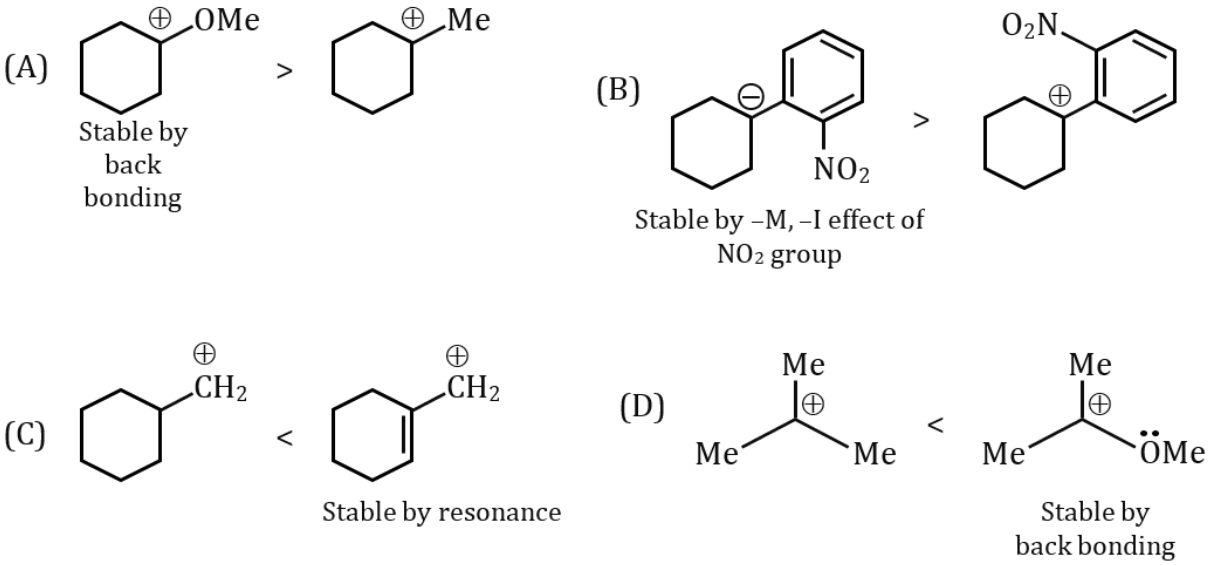
CHEMISTRY

24) 

4-Ethylcyclopent-2-en-1-ol

25) 

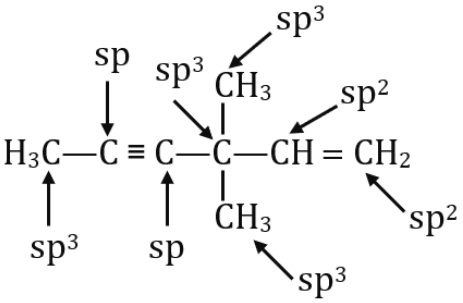
1-Bromo-2-methylbut-2-ene



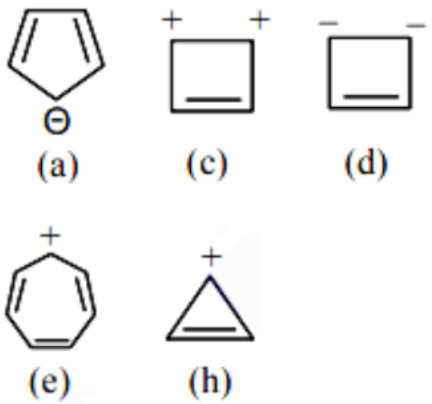
26)

27) $EtO_2C{\equiv}H$

Conjugate base of terminal alkyne will be sp hybridisation and localised. In other conjugate base, will be resonance stablised.

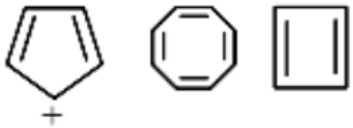


28)

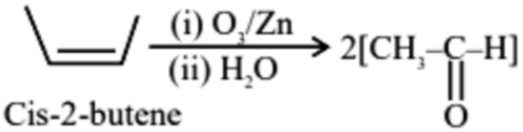


29)

a, c, d, e, h follow Huckel's rule.

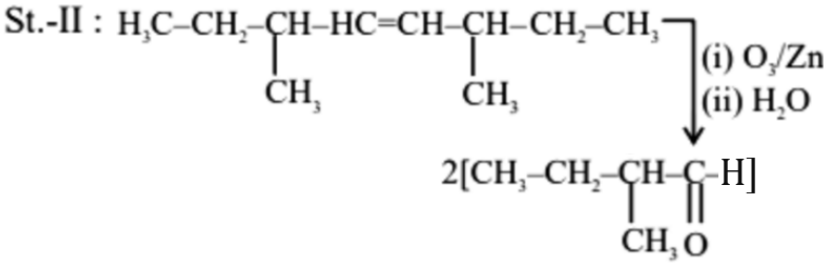


(b) (f) (g) ← These compounds do not follow Huckel's rule

St.-I : 

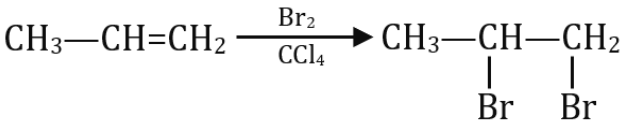
Cis-2-butene

Ethanal

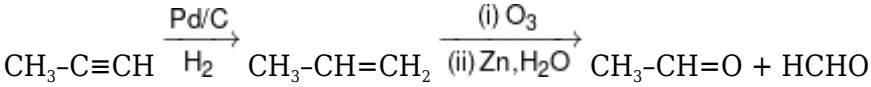
St.-II : 

30)

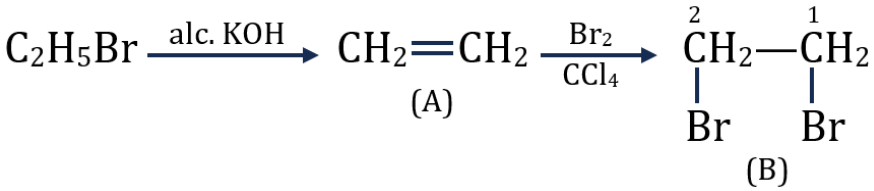
31)



Propene is most reactive towards Br_2 than others because its a electrophilic addition type of reaction.

32) 

[A] [B] [C]

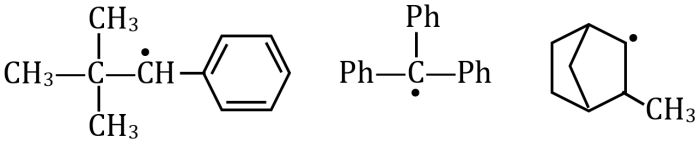


(A) (B)

1,2-dibromo ethane

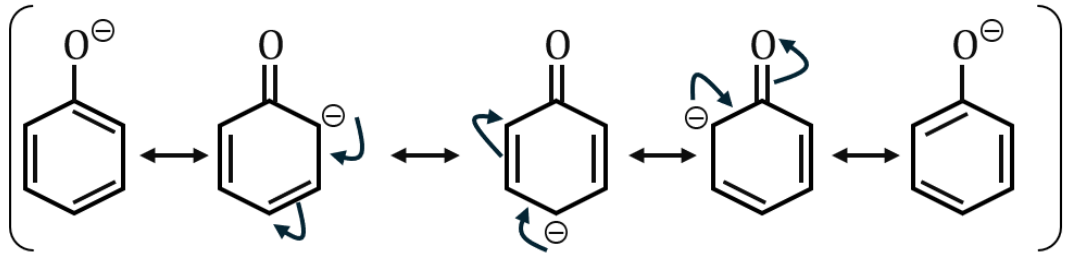
33)

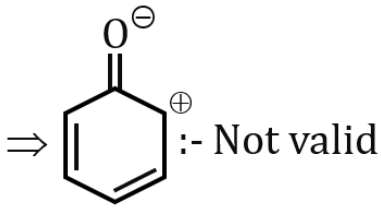
34)



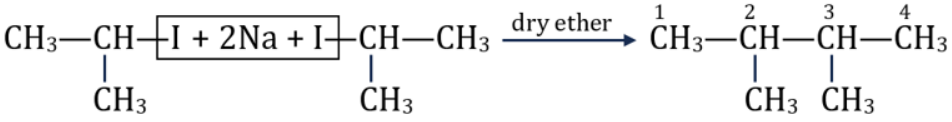
(α-H = 0) (α-H = 0) (α-H = 1)

35)



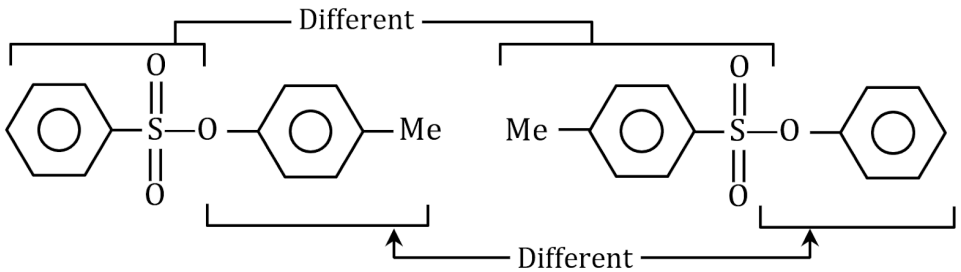


36)

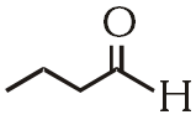
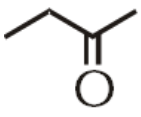


(Wurtz reaction) 2,3-dimethylbutane

37)

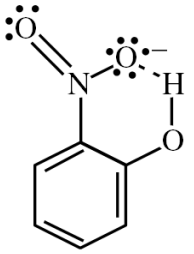


Different alkyl or aryl group/substituent attached to polyvalent functional group :- Metamerism
⇒ same bivalent functional group (sulphonate ester)
⇒ different alkyl groups on both side of central functional group
⇒ Both are metamers.

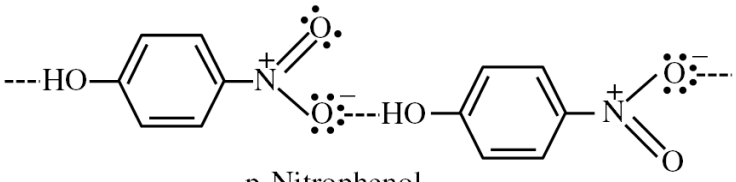
38)  and  are functional isomers and not metamers

39)

Ortho-nitrophenol: Has intramolecular hydrogen bonding, which lowers its boiling point and makes it steam volatile. Para-nitrophenol:Has intermolecular hydrogen bonding, which raises its boiling point, making it less volatile in steam. In steam distillation, compounds with lower boiling points (or those which are steam volatile) can be separated from those with higher boiling points.



o-Nitrophenol
(More volatile because of intramolecular hydrogen bonding)



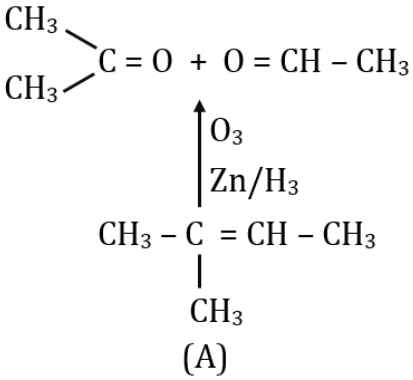
p-Nitrophenol
(Less volatile because of intramolecular hydrogen bonding)

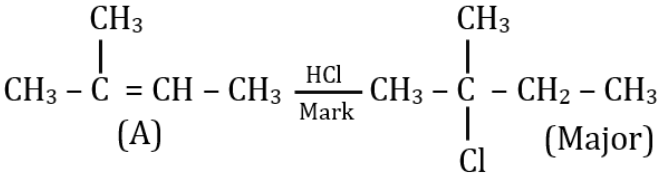
Hence, ortho-nitrophenol will distill over with steam, while para-nitrophenol will remain in the distillation flask.

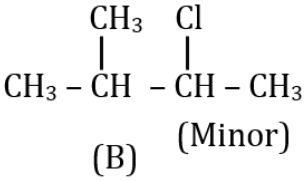
40)

The angle of rotation to achieve the anti-staggered conformation (most stable) is 180º.

41) Correct order of –I effect ⇒ $-\overset{\oplus}{N}H_3 > -C\equiv N >$ 

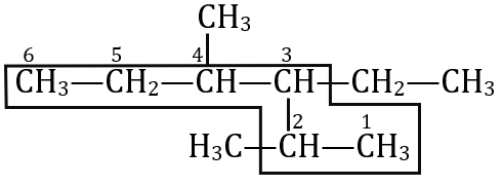






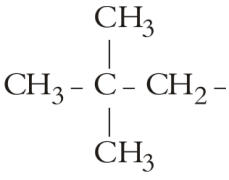
42)

43)



3–Ethyl–2,4–dimethylhexane
(Long chain as well as more no. of substituents.)

44)



[neo-pentyl not tert.]

45)

Explaining : Assertion Reason type.
Concept : This question is based on acidic nature of alkyne.
Solution :
$CH \equiv CH + NaNH_2 \rightarrow CH \equiv C^-Na^+ + NH_3 \uparrow$

$\underset{\text{(Weak Acid)}}{CH \equiv CH} + NaOH \not\rightarrow CH \equiv C^-Na^+ + \underset{\text{(Strong Acid)}}{H_2O}$

R—OH > CH ≡ CH (Acidic nature)
Final Answer : (3)

BIOLOGY

46) **Solution :** Ovum contains the haploid set of chromosomes with one of the X chromosomes. The haploid set of chromosomes in the male gamete, sperm has either the X or Y chromosome. Thus, the sex of the foetus depends on the male gamete fertilizing the ovum.
Final Answer : (4)

47) **Solution :** Second meiotic division in secondary oocyte results in the formation of a second polar body & a haploid ovum (ootid).
Final Answer : (4)

48) **Solution :** True statements are -
(ii) Egg contributes most of the cytoplasm.
(iii) Both sperm and egg contribute haploid nucleus.
Final Answer : (2)

49) **Solution :** Each testes has about 250 compartments called testicular lobules. Each lobule contains one to three highly coiled seminiferous tubules. Bulbourethral glands are paired male accessory glands.
Final Answer : (3)

50) **Solution :** The process of fertilization takes place in the ampullary part of fallopian tube. Uterus does not secrete any steroid hormones. It's the ovary which secretes the steroid

hormones.
Final Answer : (3)

51) **Solution :** The glandular tissue of each breast is divided into 15-20 mammary lobes containing clusters of cells called alveoli.
Final Answer : (2)

52) **Solution :** Graafian follicle releases secondary oocyte from the ovary by the process of ovulation. Primary oocyte within the tertiary follicle grows in size and completes its first meiotic division.
Final Answer : (2)

53) **Solution :** During pregnancy, the increased production of the levels of hormones like estrogen, progestogens, cortisol, prolactin, thyroxine in the maternal blood are essential for supporting the foetal growth, metabolic changes in the mother and maintenance of pregnancy.
Final Answer : (3)

54) **Solution :** Statement (i) and (iv) are correct. Ovary is the primary female sex organ which produces female gamete (ovum) and steroid hormones. Ovarian stroma is divided into peripheral cortex and inner medulla.
Mammary gland is a paired structure that contains glandular tissue and variable amount of tissue. It is the characteristic of all female mammals.
Final Answer : (1)

55) **Solution :** Correct statements -
(i) It produces several hormones like hCG, hPL, estrogens, progestogens etc.
(iv)It is the organ, formed in the lining of the uterus by the union of the uterine mucous membrane with the membranes of the foetus.
(v) It develops at a point of implantation and providing oxygen and nutrients for the foetus and transfer of waste products from the foetal to the maternal blood circulation.
Final Answer : (2)

56) **Solution :** Parturition, also called birth or childbirth, is a process of bringing forth a child from the uterus, or womb. Oxytocin induces uterine contraction.
Final Answer : (1)

57) **Solution :** Fertilization is the fusion of male and female gamets to form zygote. During fertilization only head of the sperm enters egg. After that polyspermy is avoided by fertilization membrane.
Final Answer : (3)

58) **Solution :** In female, Graafian follicle forms corpus luteum after ovulation. The cells of corpus luteum are called luteal cells. The cytoplasm of luteal cells have yellow granules called lutein which secrete the hormone progesterone to maintain pregnancy if fertilization takes place. In the absence of fertilization, corpus luteum degenerates and forms corpus albicans

and there is decrease in progesterone level also.
Final Answer : (2)

59) **Solution :** Clitoris is a female reproductive organ. It is homologous to penis of males. It is not remnant of penis. It is also have erectile tissue and blood supply. Penis is the copulatory organ of males.
Final Answer : (2)

60) **Solution :** Mammalian eggs are not covered by shell and yolk is not present.
Final Answer : (4)

61) **Solution :** Leydig cells, also known as interstitial cells, are found adjacent to the seminiferous tubules in the testicle. They produce testosterone in the presence of luteinizing hormone (LH).
Final Answer : (3)

62) **Solution :** Seminiferous tubules are located in the testes, which are the specific location of meiosis, and the subsequent formation of sperms. Rete testis is a network of small tubules found in the part of the testicle that carries sperm. Leydig cells or interstitial cells of Leydig produce the hormone called testosterone which contributes to male secondary sexual characters. Prepuce is the foreskin, skin surrounding and protecting the head of the penis as well as clitoris.
Final Answer : (2)

63) **Solution :** A spermatogonium divides into two primary spermatocytes. Each primary spermatocyte duplicates its DNA and subsequently undergoes meiosis I to produce two haploid secondary spermatocytes. Each of the two secondary spermatocytes further undergoes meiosis II to produce two haploid spermatids.
Final Answer : (4)

64) **Solution :** A – III; B – IV; C – V; D – II; E – I

A.	Ampulla	III.	Wider part of fallopian tube where fusion of male and female gametes takes place.
B.	Labia majora	IV.	Larger hairy folds which extend down from the mons pubis and surrounds the vaginal opening.
C.	Oviduct	V.	Also called fallopian tubes, which extend from the periphery of each ovary to the womb.
D.	Fimbriae	II.	It helps in collection of ovum after ovulation.
E.	Endometrium	I.	It undergoes cyclical changes during menstrual cycle.

Final Answer : (3)

65) **Solution :** A – III; B – I; C – IV; D – II; E – V; F – VI

A.	Primary oocyte	III.	A large number of these degenerate during the phase from puberty to birth.

B.	Secondary oocyte	I.	It is formed when oogonia starts division and temporarily arrested at prophase of meiosis I.
C.	Primary follicle	IV.	Gamete mother cell.
D.	Oogonia	II.	A large haploid cell which retains bulk of nutrient rich cytoplasm of the primary oocyte.
E.	Secondary follicle	V.	Surrounded by more layers of granulosa cells and a new theca.
F.	Graafian follicle	VI.	Rupture to release ovum from the ovary.

Final Answer : (2)

66) **Solution :** A – I; B – II; C – IV; D – III

A.	Menstrual phase	I.	Breakdown of endometrial lining of uterus along with its blood vessels which form liquid that comes out of vagina
B.	Luteal phase	II.	A temporary endocrine is formed and secretes a hormone which maintains endometrium and implantation of fertilized ovum events of pregnancy.
C.	Follicular phase	IV.	Formation of mature Graafian follicle and regeneration of endometrium of uterus.
D.	Ovulatory phase	III.	Secretion of luteinizing hormone at its maximum level and induces breakdown of mature follicle to release the female gamete.

Final Answer : (4)

67) **Solution :** A–Vas deferens B–Seminal vesicle, C–Prostate, D–Bulbourethral gland.
Final Answer : (3)

68) **Solution :** The given figure shows the sectional view of seminiferous tubule. The seminiferous tubules are the site of the germination, maturation, and transportation of the sperm cells within the male testes. Seminiferous tubules are made up of columnar sertoli cells surrounded by spermatogenic cells on the epithelial interior and stem cells exteriorly.
Final Answer : (2)

69) **Explanation:** In over 60 per cent of angiosperms, pollen grains are shed at this 2-celled stage.
Concept:
Final Answer : (4)

70) **Explanation:** The embryo sac in angiosperms is 8-nucleate but 7-celled.
Concept:
Final Answer : (4)

71) **Explanation:**
- Hilum represents the junction between ovule and funicle
- Funicle is the stalk of the ovule
- Micropyle is the small opening in the integuments
- Chalaza is the basal part of the ovule

Concept: Ovule structure

Final Answer : (1)

72) **Explanation:**
- Tapetum provides nourishment
- Exine is protective and patterned
- Filiform apparatus guides pollen tube
- Germ pore allows pollen germination

Concept: Microsporangium and embryo sac components
Final Answer : (1)

73) **Explanation:**
- MMC undergoes meiosis
- Dyad is a meiotic product (two cells)
- Tetrad = four megaspores
- Functional megaspore → embryo sac

Concept: Megasporogenesis
Final Answer : (1)

74) **Explanation:** PMC undergoes meiosis → microspore tetrad → matures into pollen grain
Concept: Pollen development
Final Answer : (1)

75) **Explanation:** The nucleus of the functional megaspore divides mitotically to form 2, then 4, then 8 nuclei
Concept: Embryo sac development
Final Answer : (1)

76) **Explanation:** Epidermis is the outermost layer, not tapetum
Concept: Microsporangium wall
Final Answer : (1)

77) **Explanation:** MMC is diploid. Hilum connects ovule to funicle (not placenta).
Concept: Ovule structure and ploidy
Final Answer : (4)

78) **Explanation:** In Rosaceae, "pollen grains maintain viability for months," not quickly lost
Concept:
Final Answer : (4)

79) **Explanation:** Micropyle is the opening through which the pollen tube enters.
Concept: Floral parts
Final Answer : (3)

80) **Explanation:** Central cell has **two polar nuclei**, not one.
Concept:

Final Answer : (3)

81) **Explanation:**
– Pollen lands on stigma (C)
– Generative cell divides during tube growth (B)
– Pollen tube enters synergid (A)
– Male gametes discharged into embryo sac (D)
Concept: Double fertilisation sequence
Final Answer : (2)

82) **Explanation:**
– Statement 1 is true: Unisexual flowers promotes cross pollination.
– Statement 2 is true: Wind (anemophily), insects (entomophily), etc. act as agents.
– Statement 3 is false: Cross-pollination increases heterozygosity.
– Statement 4 is true: Enhances variability.
Concept: Cross-pollination benefits and mechanisms.
Final Answer : (3)

83) **Explanation:**
– *Salvia* (P) is insect-pollinated: A-P
– *Maize* (Q) is wind-pollinated: B-Q
– *Vallisneria* (R) is pollinated on water surface: C-R
– *Bignonia* and similar trumpet flowers attract birds: D-S
Concept: Pollination agents and examples
Final Answer : (1)

84) **Explanation:**
– Features like inconspicuous flowers, no nectar, and abundant dry pollen are typical of a**nemophily** (wind pollination).
– Insects and birds require colourful, nectar-rich flowers
Concept: Floral adaptations to pollination agents
Final Answer : (3)

85) **Explanation:** Sticky pollen and coloured petals are traits of insect-pollinated flowers; light pollen suits wind; cleistogamy ensures self-pollination
Concept: Pollination types.
Final Answer : (1)

86) **Explanation:**
• Chasmogamous → open
• Cleistogamous → closed
• Geitonogamy → same plant
• Xenogamy → different plants
Concept:
Final Answer : (1)

87) **Explanation:** Triple fusion is necessary for the formation of endosperm. Embryo develops from syngamy, not triple fusion.
Concept: Role of triple fusion in seed development
Final Answer : (2)

88) **Explanation:** First zygote forms → endosperm develops early to nourish embryo → embryo forms → integuments harden into seed coat
Concept:
Final Answer : (1)

89) **Explanation:**
- Endosperm from PEN
- Embryo from zygote
- Seed coat from integuments
- Pericarp from ovary wall

Concept:
Final Answer : (1)

90) **Explanation:** Statement 2 is incorrect; seeds formed via apomixis are genetically identical, not variable.
Concept:
Final Answer : (2)